

Web Technologies Frontend

CSS FLEXBOX AND GRID SYSTEM

Flexbox

Flexbox is a one-dimensional layout method for arranging items in rows or columns. Items *flex* (expand) to fill additional space or shrink to fit into smaller spaces.

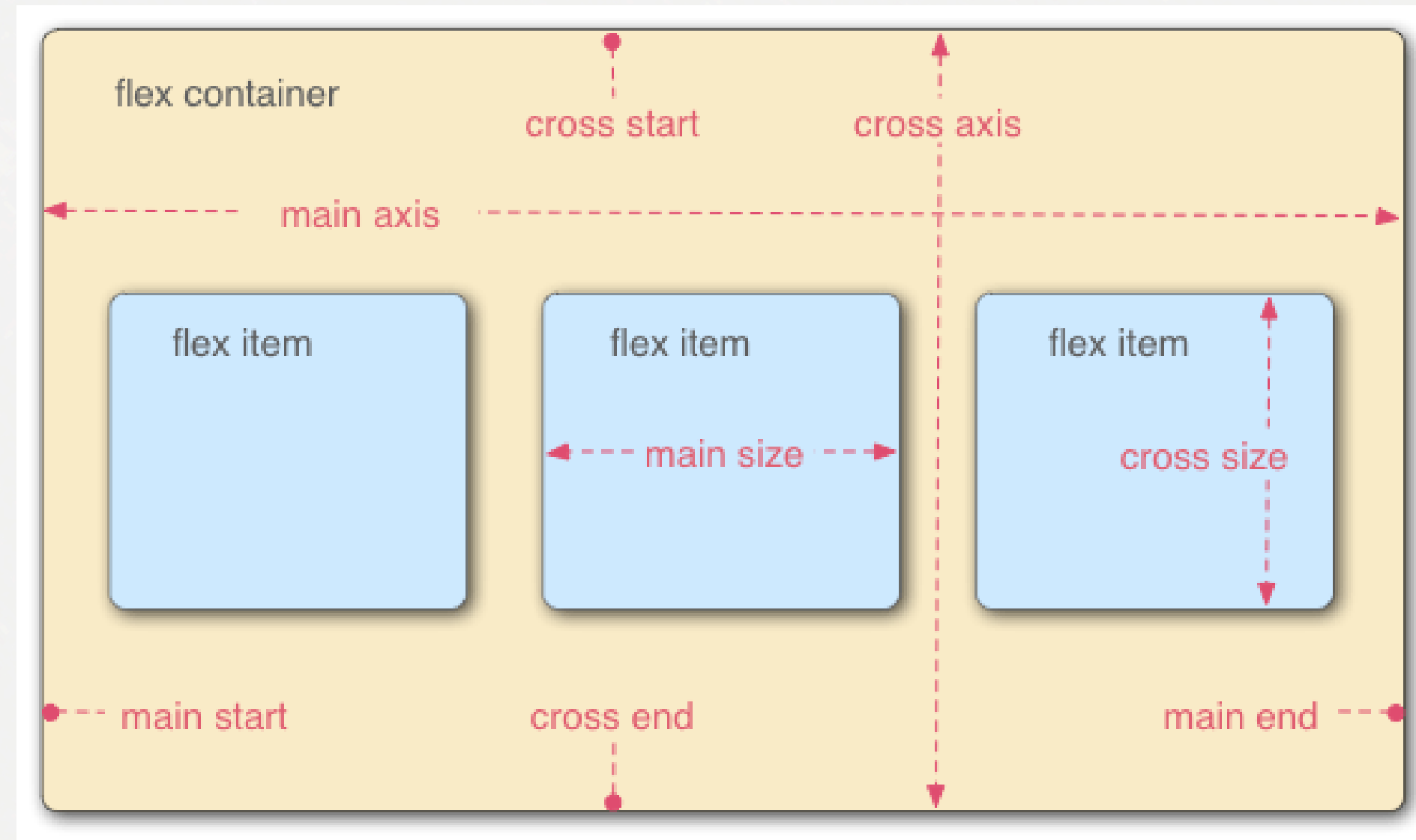
CSS flexible box layout enables you to:

- Vertically center a block of content inside its parent.
- Ensure that all children in a container occupy an equal amount of the available width/height, regardless of the available space.
- Make all columns in a multiple-column layout adopt the same height, even if they contain different amounts of content.

Flexbox terminology

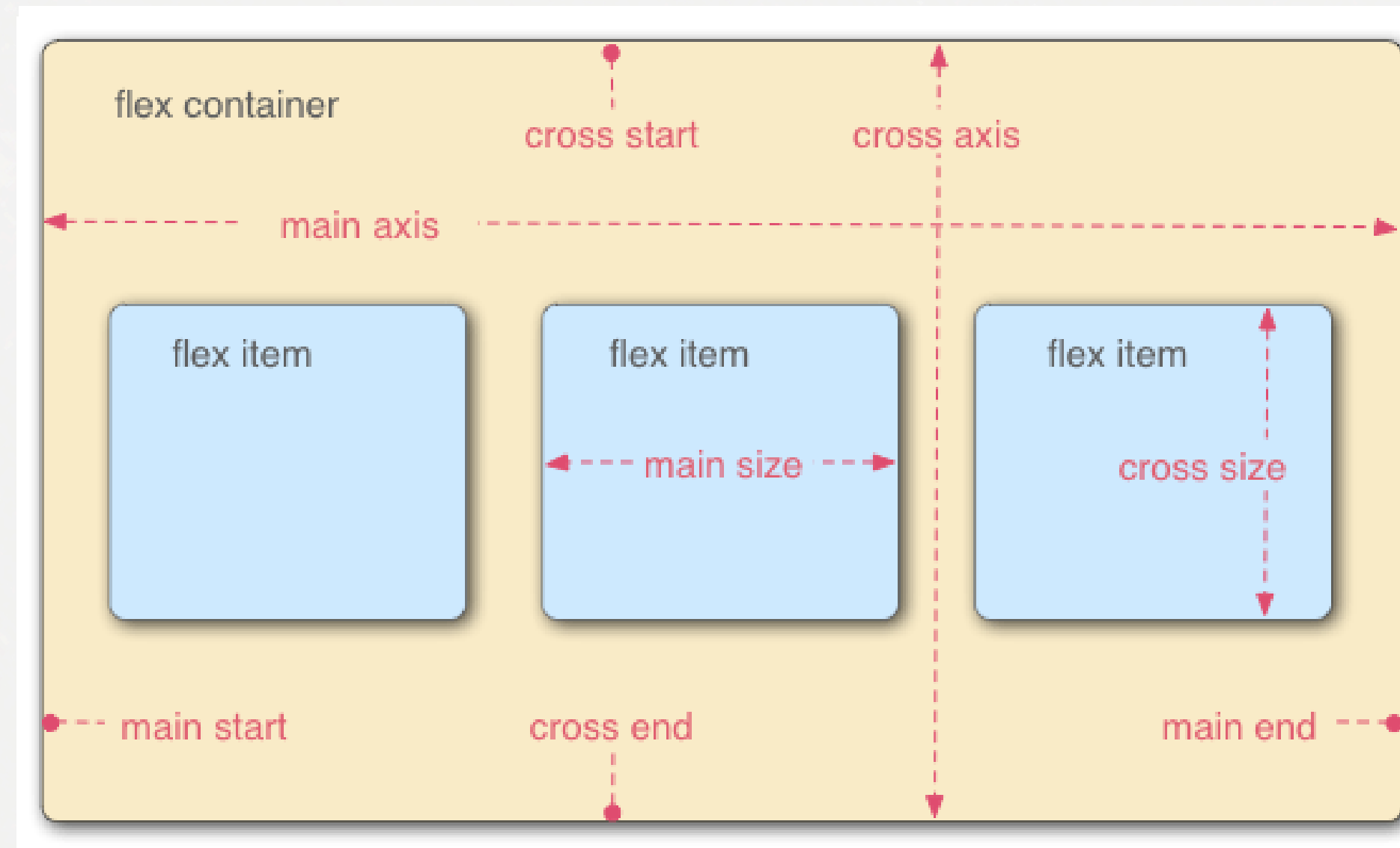
If “regular” layout is based on both block and inline flow directions, the flex layout is based on “flex-flow directions”.

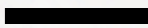
1. **Flex container:** parent element with display: flex
2. **Flex items:** direct children inside the container
3. **Main axis:** is the primary axis along which flex items are laid out
4. **Cross axis:** the axis perpendicular to the main axis



Flexbox terminology

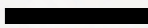
1. **main-start & main-end:** The starting and ending edges of the main axis where items are placed.
2. **main size:** The total size of flex items along the main axis (width if row, height if column).
3. **cross-start & cross-end:** The starting and ending edges of the cross axis where items are placed.
4. **cross size:** The total size of flex items along the cross axis (height if row, width if column).





Key Flexbox Properties

Level	Property	Description
Container	display	Defines a flex container; inline or block depending on the given value.
	flex-direction	Defines the main axis (row, row-reverse, column, column-reverse).
	justify-content	Aligns items on the main axis (flex-start, center, space-between, space-around, space-evenly).
	align-items	Aligns items on the cross axis (flex-start, center, stretch, baseline).
	flex-wrap	Controls wrapping of items (nowrap, wrap, wrap-reverse).

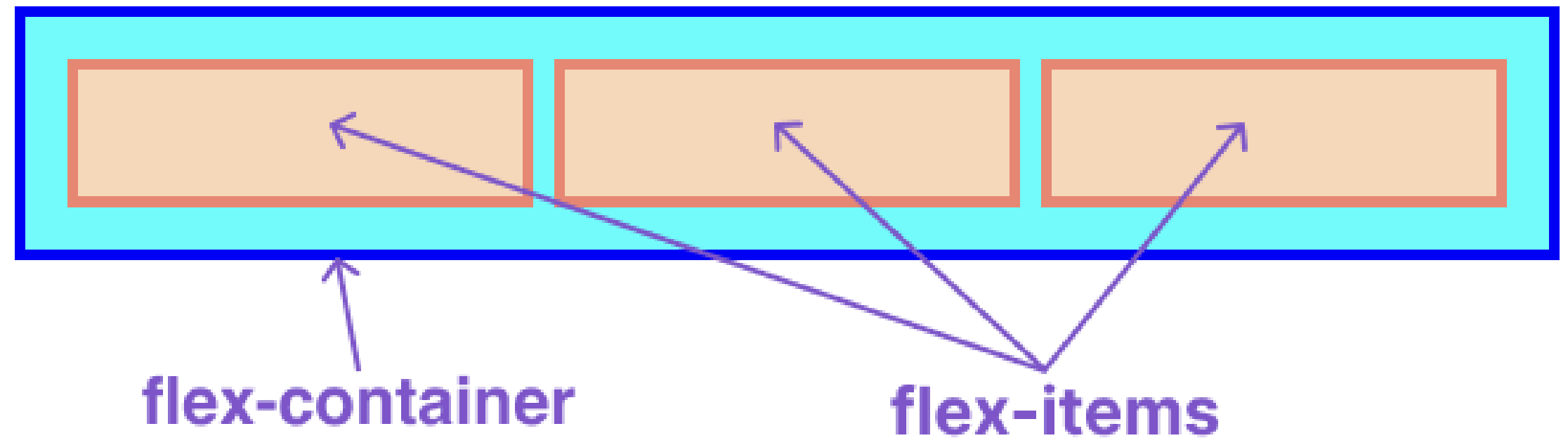


Key Flexbox Properties

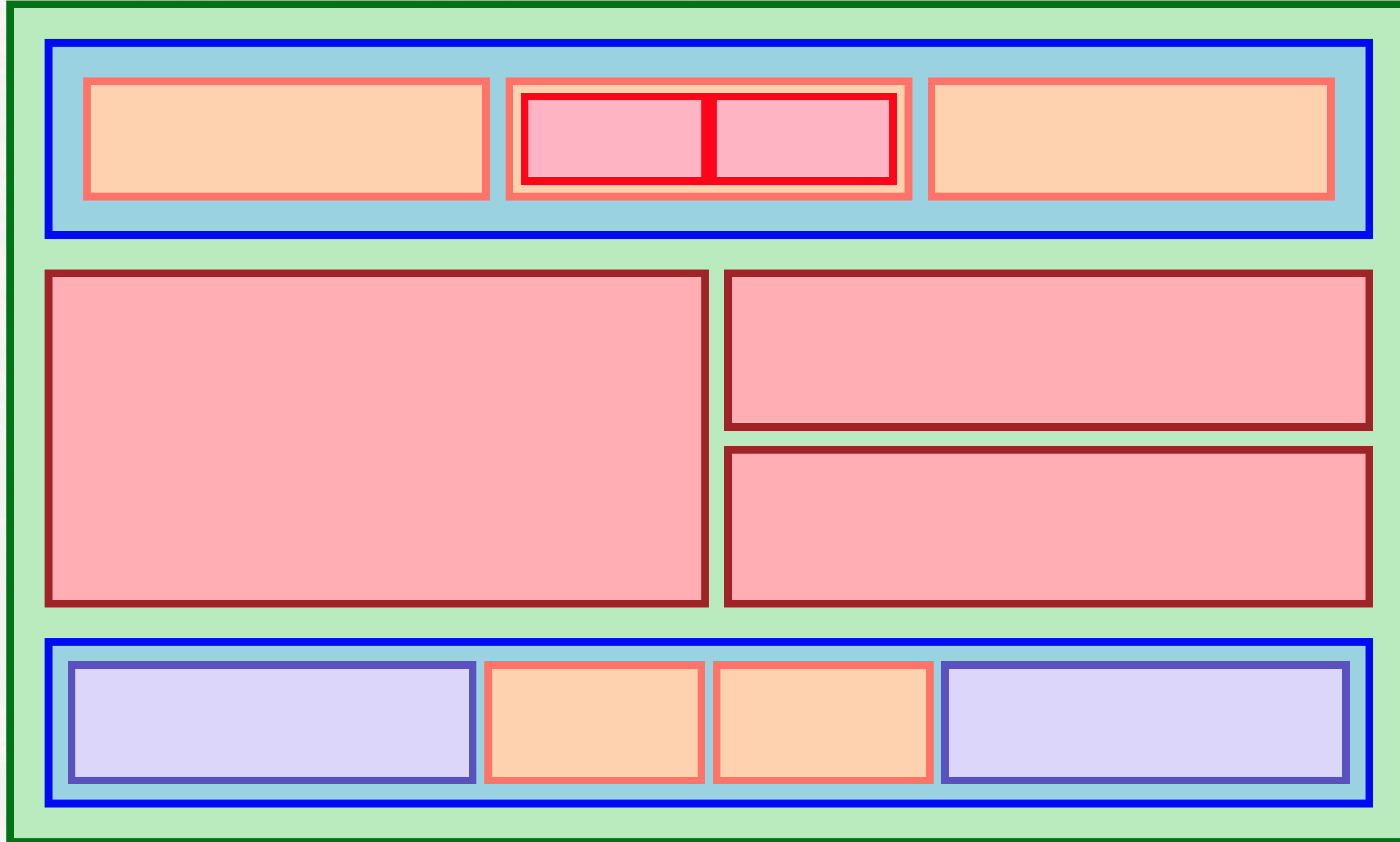
Level	Property	Description
Items	flex-grow	Defines how much an item can grow relative to others. Default 0
	flex-shrink	Defines how much an item can shrink relative to others. Default 1
	flex-basis	Defines the initial size of an item before growth/shrink. Default auto
	align-self	Overrides align-items for a single item.
	order	Controls the order in which flex items appear in the flex container. Default 0

Flexbox Example

```
.container {  
  background-color: aqua;  
  padding: 12px;  
  border: 4px solid blue;  
  display: flex;  
}  
  
.item {  
  background-color: peachpuff;  
  border: 4px solid salmon;  
  height: 48px;  
  flex: 1;  
  margin: 4px;  
}
```



Flexbox Example



Grid Layout

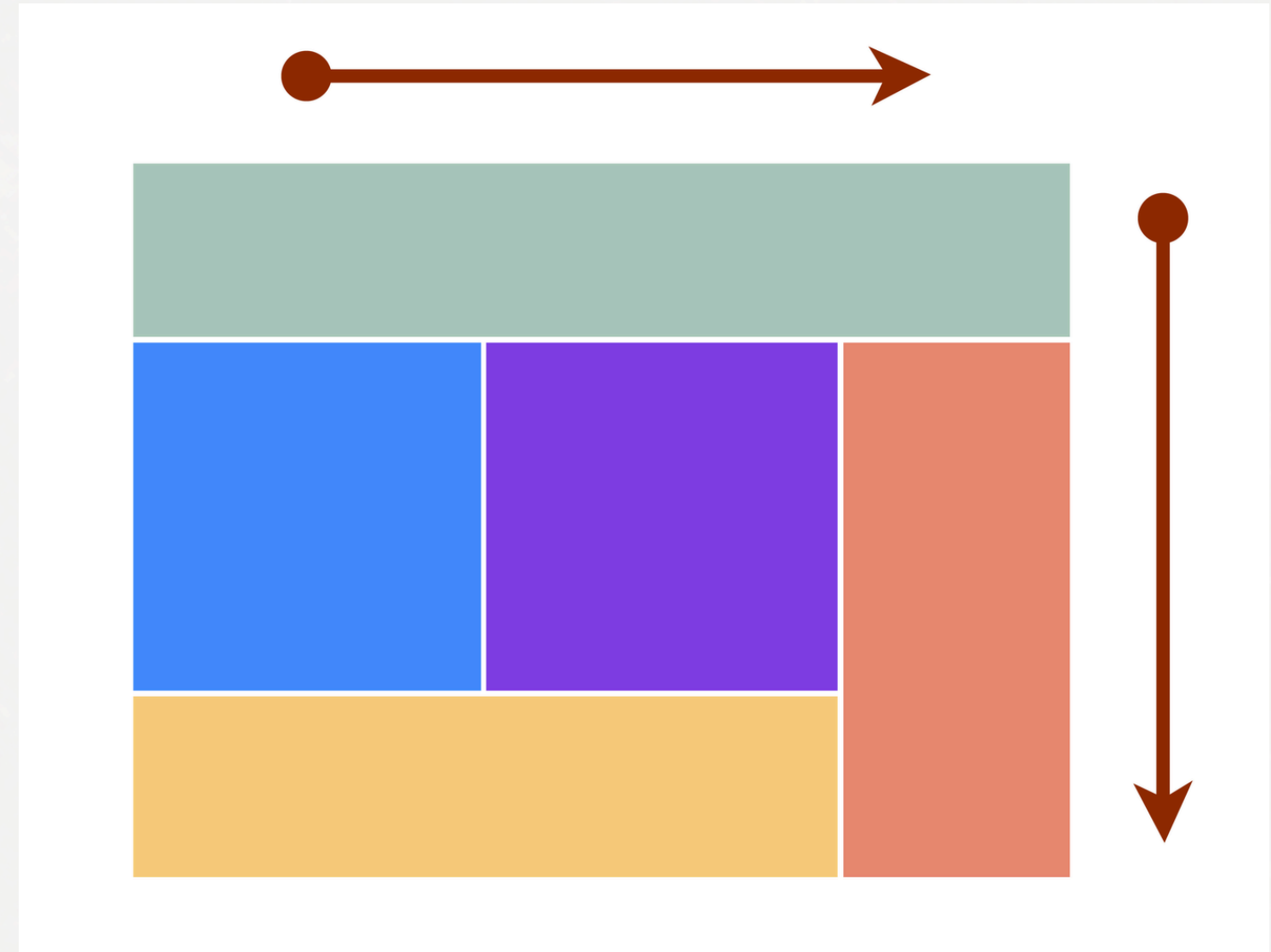
CSS Grid Layout is a two-dimensional grid-based layout system that, compared to any web layout system of the past, completely changes the way we design user interfaces.

CSS grid layout enables you to:

- Define fixed, flexible, or proportional track sizes using units like fr, auto, or %.
- Place items into specific grid cells or span them across multiple rows and columns, regardless of document order.
- Build complex page layouts (dashboards, galleries, templates) with consistent alignment and spacing.

Grid terminology

1. **Grid Container:** parent with display: grid
2. **Grid Items:** children inside the grid
3. **Grid Track:** rows and columns
4. **Grid Area:** rectangular space of multiple cells
5. **Grid Line:** dividing lines between cells
6. **Grid Cell:** a single “unit” of the grid.

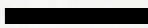


Special Units & Functions

Level	Property	Description
Fractional Units	fr	Represents a fraction of available space in the grid container. Example: 1fr 2fr.
Sizing Units	auto	Size adjusts automatically to fit content.
	min-content	Tracks shrink to the smallest size needed for content.
	max-content	Tracks expand to the largest size needed for content.
Sizing Functions	minmax (min, max)	Defines a track that resizes between a min and max value.
	fit-content (value)	Track grows to fit content up to a maximum value.
Repetition	repeat (n, value)	Repeats track definitions a specified number of times. Example: repeat(3, 1fr).

Key Grid Properties

Level	Property	Description
Container	grid-template-columns	Defines the number and width of columns.
	grid-template-rows	Defines the number and height of rows.
	gap / row-gap / column-gap	Controls spacing between grid tracks.
	align-items	Aligns items vertically within their cells.
	justify-items	Aligns items horizontally within their cells.
	align-content	Aligns the entire grid along the column axis inside the container.
	justify-content	Aligns the entire grid along the row axis inside the container.



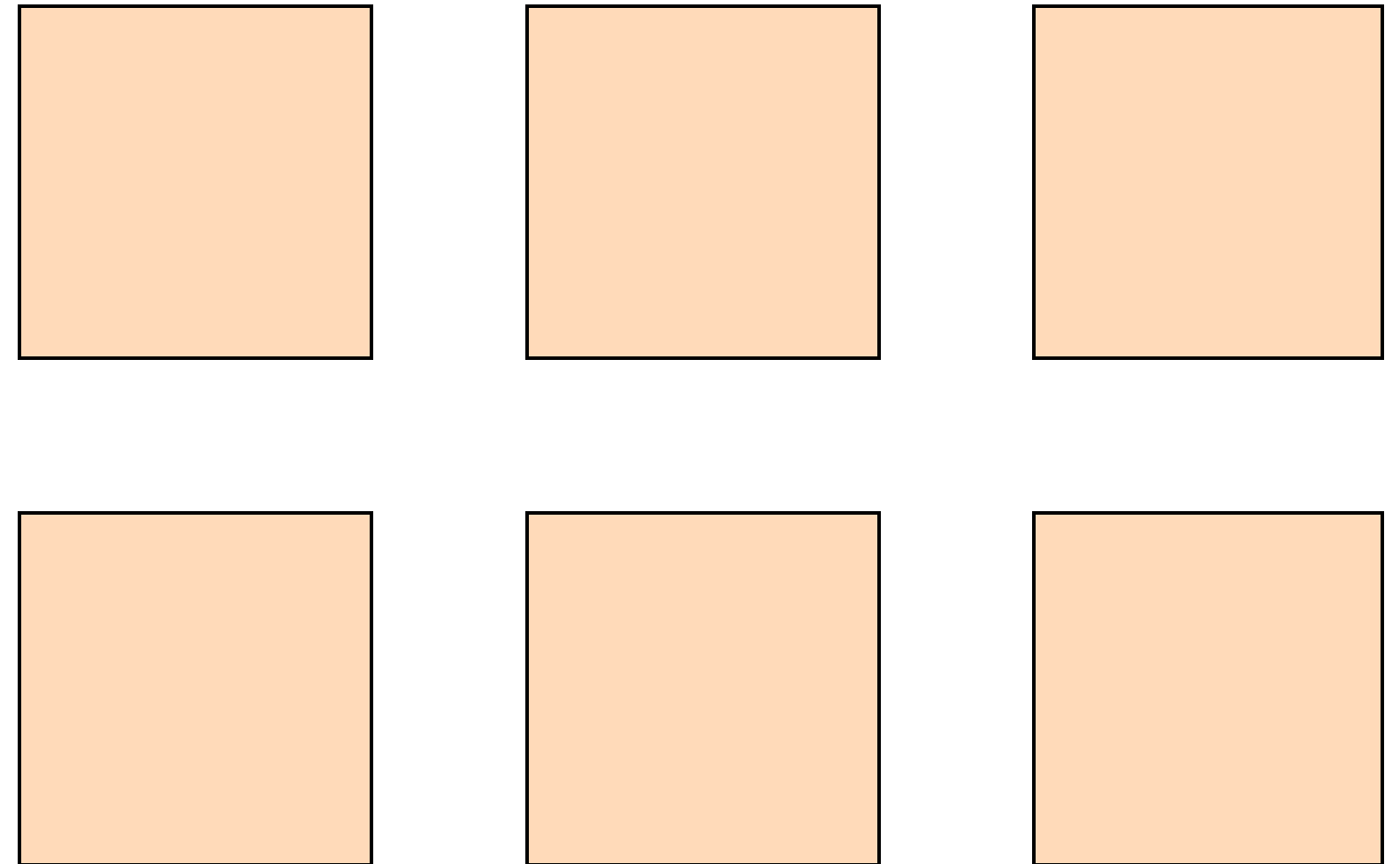
Key Grid Properties

Level	Property	Description
Items	grid-column	Specifies the column(s) an item should occupy.
	grid-row	Specifies the row(s) an item should occupy.
	grid-area	Defines an item's position by both row and column (can name areas too).
	justify-self	Aligns a single item horizontally inside its cell.
	align-self	Aligns a single item vertically inside its cell.

Grid Example

```
.grid-container {  
  display: grid;  
  grid-template-columns: 75px 75px 75px;  
  grid-template-rows: 75px 75px;  
  gap: 32px;  
}
```

```
.card {  
  background-color: peachpuff;  
  border: 1px solid black;  
}
```



Flexbox vs Grid

Feature	Flexbox (1D)	Grid (2D)
Layout direction	Row OR Column	Rows AND Columns
Use case	Navigation bars, simple alignment	Complex page layouts
Alignment power	Excellent	Excellent
Responsiveness	Easy with wrap	Easy with auto-fit/fill

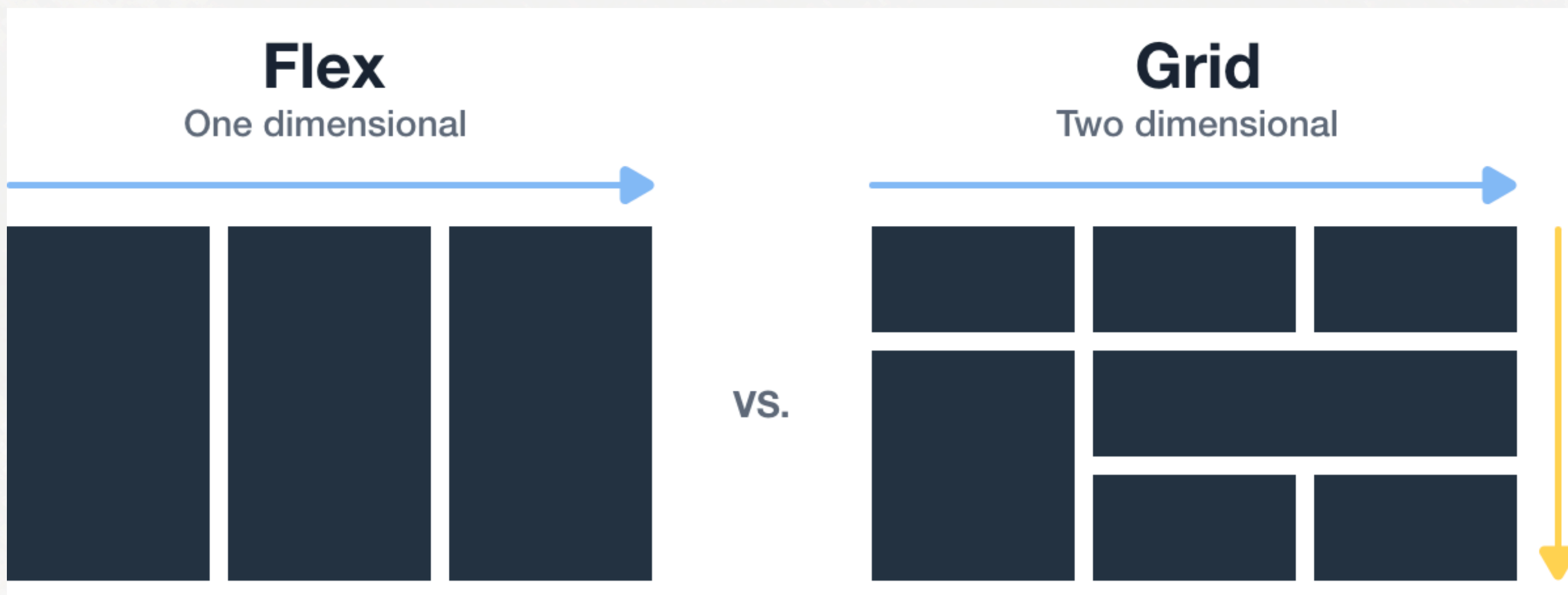
When to use which?

Use Flexbox:

- Navbars, menus, buttons, form controls
- One-direction layouts

Use Grid:

- Full page layout
- Dashboards, galleries, complex responsive designs



Combining Flexbox and Grid

- Grid can do things Flexbox can't do.
- Flexbox can do things Grid can't do.
- They can work together: a grid item can be a flexbox container. A flex item can be a grid container.

Use case:

- Grid for large-scale layout
- Flexbox for aligning content inside grid cells

THANK YOU FOR LISTENING

Any questions?

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